

AI in Blockchain

“Blockchain is a shared, immutable digital ledger that enables recording of transactions and the tracking of assets within a business network and providing a single source of truth” (Susnjara & Smalley, 2025). It operates as a decentralized distributed database; the data is stored across multiple computers, making it resistant to tampering or unauthorized changes.

In blockchain technology, each transaction is validated through a consensus mechanism, a process where multiple participants in the network must agree that the transaction is valid before it is approved, then it is grouped into blocks that are linked together in chronological order, forming a secure and transparent chain. This structure is key to guaranteeing data integrity, since altering any single block would require changing every block that follows it across the entire network. Blockchain is ideal for applications like cryptocurrencies.

AI agents with wallets for automated B2B payments

In business to business (B2B) world, most payment processes rely on an individual pressing the “approve” button before funds are sent. AI agents are starting to change this by handling

funds and payments on their own while adhering to budget limitations, spending rules, and approval policies. This helps reduce staffing costs and speed up payment cycles.

Right now, SaaS companies that automate payment to suppliers and logistic companies are some of the main users of these AI agents because their payment volumes are high and repetitive, so they benefit from automation. Even though this new technology is changing the field, companies are not removing humans completely, following the human-in-the-loop framework, so they can monitor and intervene in the agent's decisions and reduce risks.

AI Smart contracts

A smart contract is a software program stored on the blockchain, fixing terms and conditions of the agreement in its code and using if-then logic statements for execution. For example, if the payment is made, the product is delivered.

Smart contracts have structural limitations, since their logic does not change, even when the world around them does. AI smart contracts are the solution to this problem. They integrate machine learning, AI agents, and real-time data by introducing adaptability. These systems can respond to changing inputs, optimize outcomes, and make more complex decisions.

A real-world example would be Decentralized Finance (DeFi), where AI smart contracts can dynamically manage liquidity, pricing and risk.

Transparency / Explainability

Transparent AI refers to systems where the decision-making process, used algorithms, and data is understandable and accessible. In the blockchain field, sometime transparency conflicts with privacy or security. For example, making a fully transparent system can expose sensitive data. Balancing transparency with privacy is key to AI systems in blockchain and is one of the main challenges because it requires careful planning.

Some of the benefits of making a transparent system are that builds trust. For example, a healthcare provider can share blockchain-logged decision-making data with patients, and ensure that they understand how decisions are made, building trust, and improving patient satisfaction.

Bias and fairness

Bias is one of the most discussed ethical considerations in AI. There are common forms of bias, like data bias, algorithmic bias, and deployment bias. This problem is even harder to address in blockchain, since smart contracts are automatically executed once deployed. An example of this would be a decentralized lending platform that uses AI to approve loans, which could inadvertently favor certain demographics if the training data is skewed. This is a real concern in DeFi lending platforms, since research has found racial and gender bias in peer-to-peer lending algorithms, even when these protected attributes are excluded as inputs. This happens because the AI can indirectly infer this information from other data patterns.

Some mitigation strategies are using a diverse dataset that reflects real-world scenarios and conducting regular algorithmic audits.

Future trends

In some predictions in 5 to 10 years, AI agent will take over 80% of DeFi transactions by 2030, moving from simple automation to autonomous decision-making. Humans will remain the ultimate owner and architects of strategies, but day to day execution of certain tasks like yield farming or risk management will be handled by AI.

This future comes with risks. For example trust, when an AI model makes a decision users cannot verify if the model worked correctly and even sometimes they cannot tell why that decision was made. This is a small problem for a chatbot or generative AI, but a big problem when we talk about blockchain and money. In the future, the biggest challenge will be to construct transparent AI systems that people can trust.

In conclusion, AI in blockchain has a big challenge ahead, and it's precisely transparency and being able to trust it. We already know that AI systems fail continuously, and they also have this "black box" property, so trusting them blindly to handle such an important part of our lives is a big deal.

This would be an industry I would like to work in, especially because of my interest in long-term investing. I think it could be very interesting to gain knowledge about blockchain.

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